Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2016

BIOLOGY **9648/03**

Applications Paper and Planning Question **26 Aug 2016**

PAPER 3 **2 hours**

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
TOTAL	**72**

This document consists of **16** printed pages.

1 The first restriction enzyme was discovered in *Escherichia coli* in the 1960s, and thousands more have been discovered since. These enzymes opened the path to powerful research tools which scientists use to sequence genomes through the process of genetic engineering.

(a) Compare the roles of restriction enzymes occurring naturally in bacteria and those used in genetic engineering.

[2]

(b) A gene, *URA3*, was discovered to be a potential marker gene for genetic engineering. This gene is obtained from yeast and codes for orotidine 5'-phosphate decarboxylase, which converts 5-fluoroorotic acid (5-FOA) into the toxic compound 5-fluorouracil, so any cells carrying the *URA3* gene will not survive in the presence of 5-FOA.

In an investigation of the effectiveness of *URA3* as a marker gene, researchers created a plasmid vector shown in Fig. 1.1.

A eukaryotic gene was inserted into this plasmid and the mixture was added to competent *Escherichia coli* for transformation to take place. These bacteria were then grown on agar plates containing X-gal and 5-FOA.

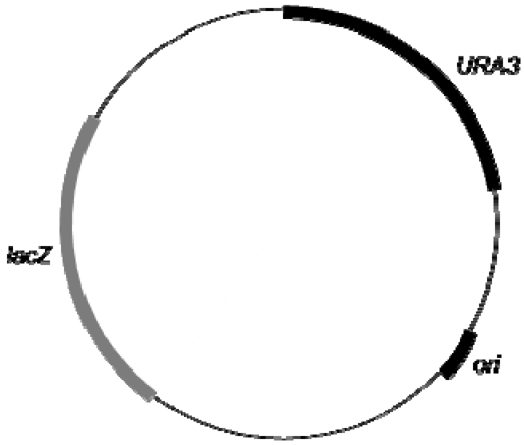

Fig. 1.1

(i) The researchers observed that recombinant bacteria appeared as blue colonies on agar plates containing X-gal and 5-FOA. Using this information, draw a cross (**X**) on the plasmid in Fig. 1.1 to indicate where the eukaryotic gene was inserted. [1]

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(ii) Account for the observations stated in **b(i)**.

[4]

(iii) State the purpose of the *lacZ* gene.

[1]

(iv) Explain why the bacteria containing the recombinant plasmid should be further subjected to gene probing.

[3]

(c) The differences between the formation of genomic and cDNA libraries are due to the original source of genetic material used. In genomic libraries, the genetic material used is the genome of a haploid cell while in cDNA libraries, mRNA from a specialised cell is extracted.

(i) Describe two differences in the processes involved in the formation of these libraries.

[2]

(ii) State two applications of cDNA libraries.

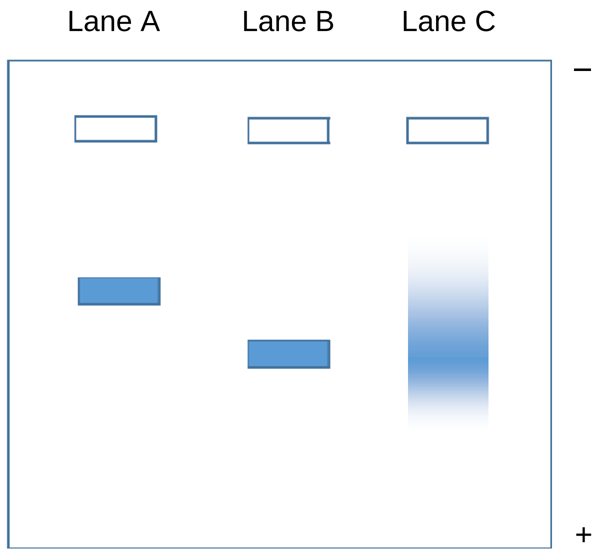
[2]

[Total: 15m]

For Examiner's Use

2 Gel electrophoresis is a technique that is widely used in molecular biology to separate DNA fragments based on molecular mass. The molecular mass of DNA fragments may differ due to the number of nucleotides present in the DNA fragment as well as chemical modifications made to the DNA.

Fig. 2.1 shows the gel electrophoregram of the DNA fragments taken from three sources. DNA fragments from all three samples are known to have equal lengths.



Key

Lane A: DNA fragments comprising insulin gene isolated from liver cells

Lane B: DNA fragments comprising insulin gene isolated from Islets of Langerhans

Lane C: Linear DNA isolated from many lambda phages

Fig. 2.1

(a) Describe the role of the buffer solution in gel electrophoresis.

....

....

....

.... [2]

(b) (i) Describe and explain the position of the band in lane A with respect to the band in lane B.

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.... [3]

(ii) Fig. 2.2 shows the DNA of a lambda phage.

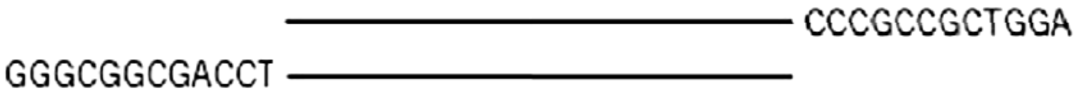

Fig. 2.2

Describe and explain the band pattern of the lambda phage DNA in lane C.

[4]

Cyclopselemia is a new genetic disease recently discovered in a small town called X-mansion. A patient suffering from cyclopselemia has red eyes and cannot control the amount of light entering his eyes. They have to constantly wear sun glasses. Scientists used information from a pedigree chart as well as RFLP analyses of the individuals in the pedigree chart to determine the allele that is responsible for this genetic disease.

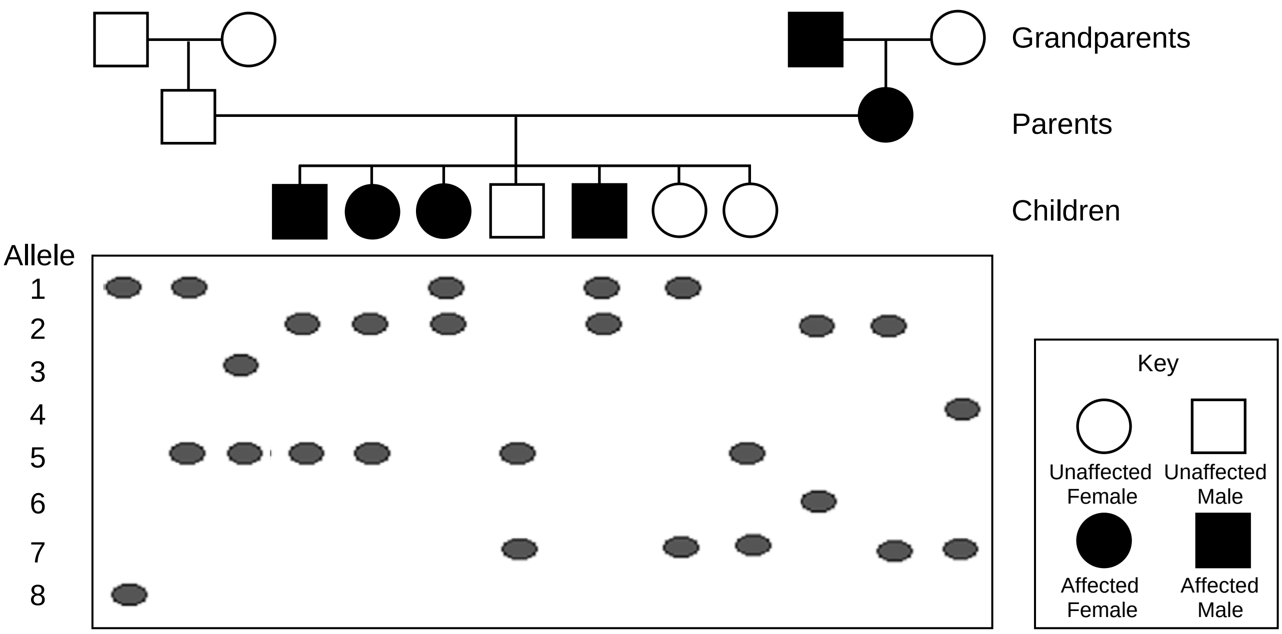

Fig. 2.3

(c) Outline the process that can be used to determine the presence of cyclopselemia in a foetus.

[5]

[Total: 14m]

3 Embryonic stem cells, which are derived from the inner cell mass of mammalian blastocysts, are used in research to understand disease development and develop future cell-based therapies for currently untreatable diseases.

In 2006, a team of scientists led by Shinya Yamanaka discovered that differentiated cells can be isolated and modified to an embryonic-like state. These cells, subsequently known as induced pluripotent stem cells (iPSCs), have been hailed as an effective replacement for human embryonic stem cells for its usefulness in regenerative medicine. Fig. 3.1 shows a potential application of human iPSCs in gene therapy.

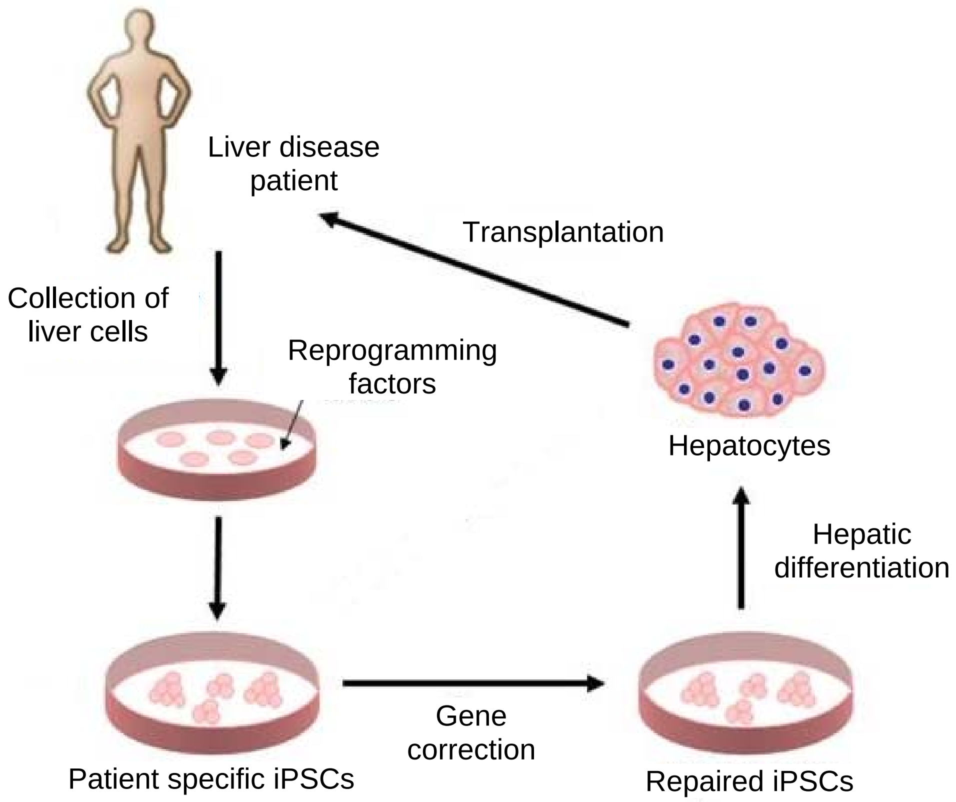

Fig. 3.1

(a) (i) State two similarities in the features of embryonic stem cells and iPSCs.

....

....

....

.... [2]

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(ii) State and explain two advantages of using iPSCs over embryonic stem cells in gene therapy.

.... [4]

Yamanaka's research team studied 24 genes expressed by embryonic stem cells in mice to identify genes that can induce pluripotency. They discovered that four genes, notably *Oct4*, *Sox2*, *Klf4* and *c-Myc*, encode transcription factors which could be used to reprogramme differentiated cells to form iPSCs. These genes can be introduced into differentiated somatic cells via viral transduction or the introduction of non-integrating plasmids. This process is shown in Fig. 3.2.

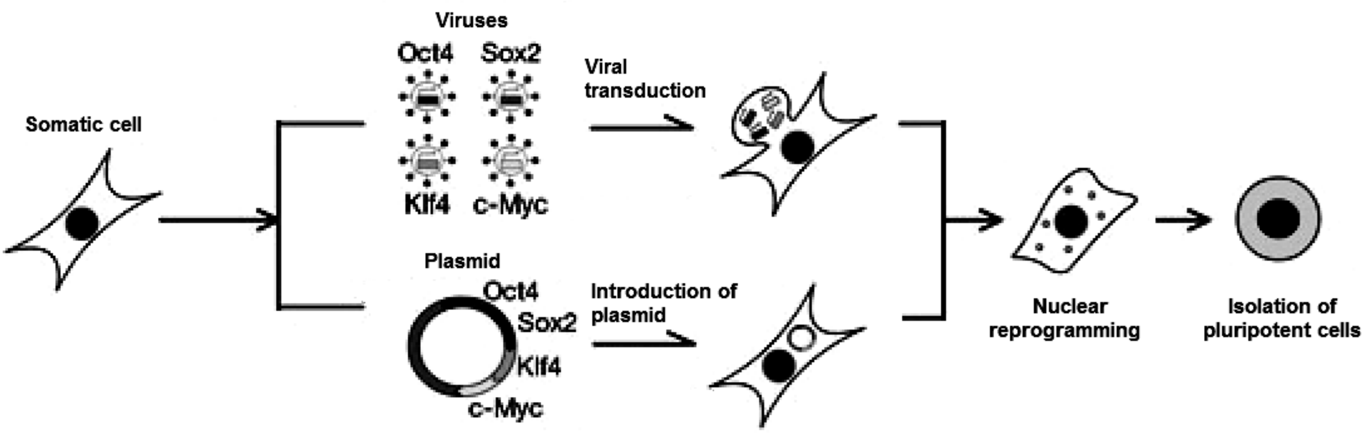

Fig. 3.2

(b) (i) Suggest why it is necessary to introduce these four genes into somatic cells in iPSC formation.

.... [1]

(ii) Describe two advantages and one disadvantage of introducing the four genes via non-integrating plasmids over viral transduction using retroviruses.

[3]

(iii) Before iPSCs are stimulated to undergo cellular differentiation, the introduced genes must be removed from the cells. Suggest why this process is necessary.

[1]

[Total: 11m]

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4 Planning question

The visking tubing is a semi-permeable membrane which allows smaller molecules such as water and glucose to pass through, but not larger molecules like sucrose and proteins. The tubing is to be supported in a boiling tube as shown in Fig. 4.1.

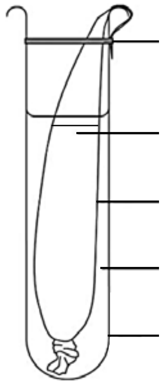

Fig. 4.1

The presence of glucose can be detected using the Benedict's test for reducing sugars.

Using this information and your own knowledge, design an experiment to investigate the effect of glucose concentration on its rate of diffusion across the visking tubing.

You must use:
- 5 pieces of visking tubing (each knotted at one end, and open at the other), soaked in a beaker of distilled water
- 5 boiling tubes and 5 rubber bands
- 2.0 moldm^{-3} glucose solution
- Benedict's solution
- weighing balance and weighing paper
- filter paper and funnel
- distilled water
- thermostatically-controlled water bath

You should select from the following apparatus:
- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, glass rods
- tripod stand, bunsen burner and lighter
- test-tube holder
- stopwatch
- syringes

Your plan should:
- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12m]

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5 Free-response Question

Write your answers to this question on the separate writing paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Describe the steps and advantages of plant tissue culture. [8]

(b) Discuss the benefits and ethical issues related to the use of a named genetically-modified animal. [6]

(c) Discuss the goals and benefits of the human genome project. [6]

[Total: 20m]